

EXPERIMENT LOGGING

WAFT | Recording Scientific Data

Why Log Experiments?

WAFT produces scientific data. Proper logging ensures:

- **Reproducibility** — Others can replicate your results
- **Analysis** — You can study what happened
- **Publication** — Data for research papers
- **Learning** — Understand what works

What Gets Logged

Automatic

- All Flight Recorder events
- Gym evaluations
- Mutations applied
- Fitness scores

Manual

- Experiment hypotheses
- Observations
- Configuration changes
- Conclusions

Experiment Structure

```
from waft.experiments import Experiment
```

```
exp = Experiment(  
    name="mutation_rate_study",  
    hypothesis="Higher mutation rates escape local optima faster",  
    variables={  
        "independent": ["mutation_rate"],  
        "dependent": ["fitness", "generations_to_plateau"],  
        "controlled": ["population_size", "gym_scenarios"],  
    },  
)
```

Running Experiments

```
# Define conditions
conditions = [
    {"mutation_rate": 0.05},
    {"mutation_rate": 0.10},
    {"mutation_rate": 0.15},
    {"mutation_rate": 0.20},
]

# Run each condition
for condition in conditions:
    exp.run_trial(
        config=condition,
        generations=100,
        replicates=3, # Run 3 times each
    )

# Analyze results
results = exp.analyze()
exp.save_results("results/mutation_study.json")
```

Logging Observations

```
# Manual observations
exp.log_observation(
    "Generation 45: Interesting behavior emerged. Agent started "
    "generating self-referential statements.",
    tags=["emergence", "unexpected"],
)

# Structured notes
exp.log_note(
    title="Plateau Detected",
    content="Fitness stuck at 0.73 for 10 generations",
    generation=55,
    action_taken="Increased mutation rate to 0.15",
)
```

CLI Commands

```
# Create experiment
waft experiment create "mutation_study" \
    --hypothesis "Higher mutation escapes optima"

# Log observation
waft experiment log "Interesting behavior at gen 45"

# View experiment
waft experiment show mutation_study

# Export data
waft experiment export mutation_study --format csv
```

Best Practices

1. **State hypothesis upfront** — What do you expect?
2. **Control variables** — Change one thing at a time
3. **Multiple replicates** — Run each condition 3+ times

4. **Log everything** — Better too much than too little
5. **Analyze promptly** — Review results while context is fresh

Report Generation

```
# Generate experiment report
```

```
waft experiment report mutation_study --format pdf
```

```
# Include in publication
```

```
waft experiment export mutation_study --format latex
```

EXPERIMENT LOGGING | Science Requires Records